

Data cleaning

INTRODUCTION TO KNIME



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Data cleaning



Filter columns

#	Name	Department	Salary	Favorite food
1	Jenny	Sales	?	Pizza
2	Alex	Finance	22000	Sushi
3	Taylor	R&D	28000	Tacos
4	Alex	Finance	22000	Sushi
5	Sam	R&D	23000	Salad

Filter columns

					Filter out
#	Name	Department	Salary	Favorite food	
1	Jenny	Sales	?	Pizza	
2	Alex	Finance	22000	Sushi	
3	Taylor	R&D	28000	Tacos	
4	Alex	Finance	22000	Sushi	
5	Sam	R&D	23000	Salad	

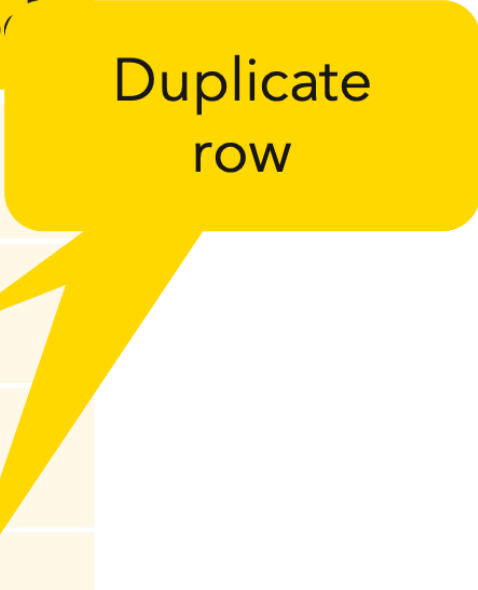
Filter rows

#	Name	Department	Salary	Favorite food
1	Jenny	Sales	?	Pizza
2	Alex	Finance	22000	Sushi
3	Taylor	R&D	28000	Tacos
4	Alex	Finance	22000	Sushi
5	Sam	R&D	23000	Salad

Employees from R&D

Filter duplicates

#	Name	Department	Salary	Favorite food
1	Jenny	Sales	?	Pizza
2	Alex	Finance	22000	Sushi
3	Taylor	R&D	28000	Tacos
4	Alex	Finance	22000	Sushi
5	Sam	R&D	23000	Salad



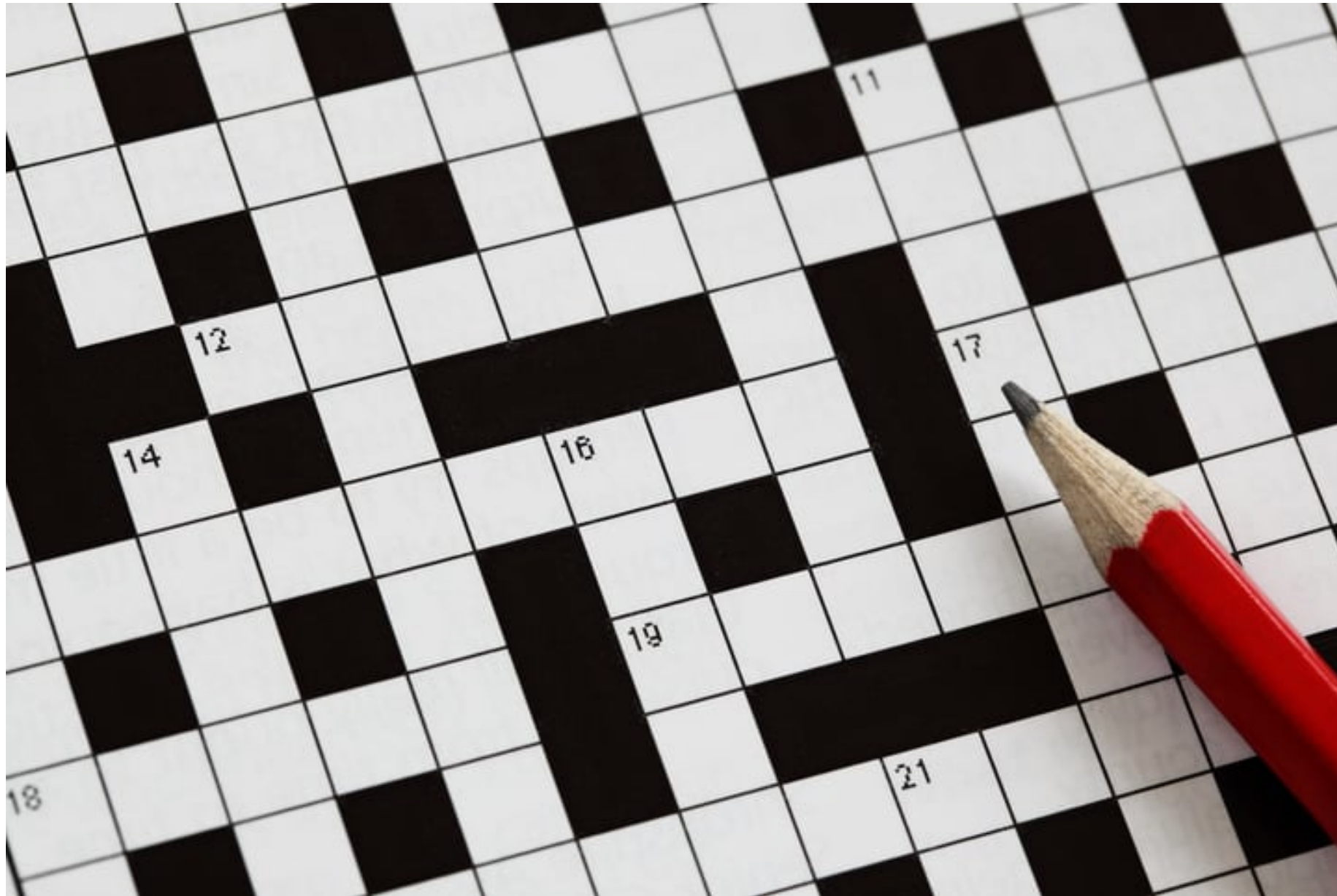
Duplicate row

Handle missing values

#	Name	Department	Salary	Favorite food
1	Jenny	Sales	?	Pizza
2	Alex	Finance	22000	Sushi
3	Taylor	R&D	28000	Tacos
4	Alex	Finance	22000	Sushi
5	Sam	R&D	23000	Salad

Missing value

Handle missing values



Data types

#	Name	Salary	Birthday	Remote
1	Jenny	25000	12/12/1990	TRUE
2	Alex	22000	04/03/1998	FALSE
3	Taylor	28000	11/02/1987	FALSE
4	Alex	22000	04/03/1998	FALSE
5	Sam	23000	02/07/1989	TRUE

Data types

String

#	Name	Salary	Birthday	Remote
1	Jenny	25000	12/12/1990	TRUE
2	Alex	22000	04/03/1998	FALSE
3	Taylor	28000	11/02/1987	FALSE
4	Alex	22000	04/03/1998	FALSE
5	Sam	23000	02/07/1989	TRUE

Data types

String		Number			
#	Name	Salary	Birthday	Remote	
1	Jenny	25000	12/12/1990	TRUE	
2	Alex	22000	04/03/1998	FALSE	
3	Taylor	28000	11/02/1987	FALSE	
4	Alex	22000	04/03/1998	FALSE	
5	Sam	23000	02/07/1989	TRUE	

Data types

String		Number	Date&Time	
#	Name	Salary	Birthday	Remote
1	Jenny	25000	12/12/1990	TRUE
2	Alex	22000	04/03/1998	FALSE
3	Taylor	28000	11/02/1987	FALSE
4	Alex	22000	04/03/1998	FALSE
5	Sam	23000	02/07/1989	TRUE

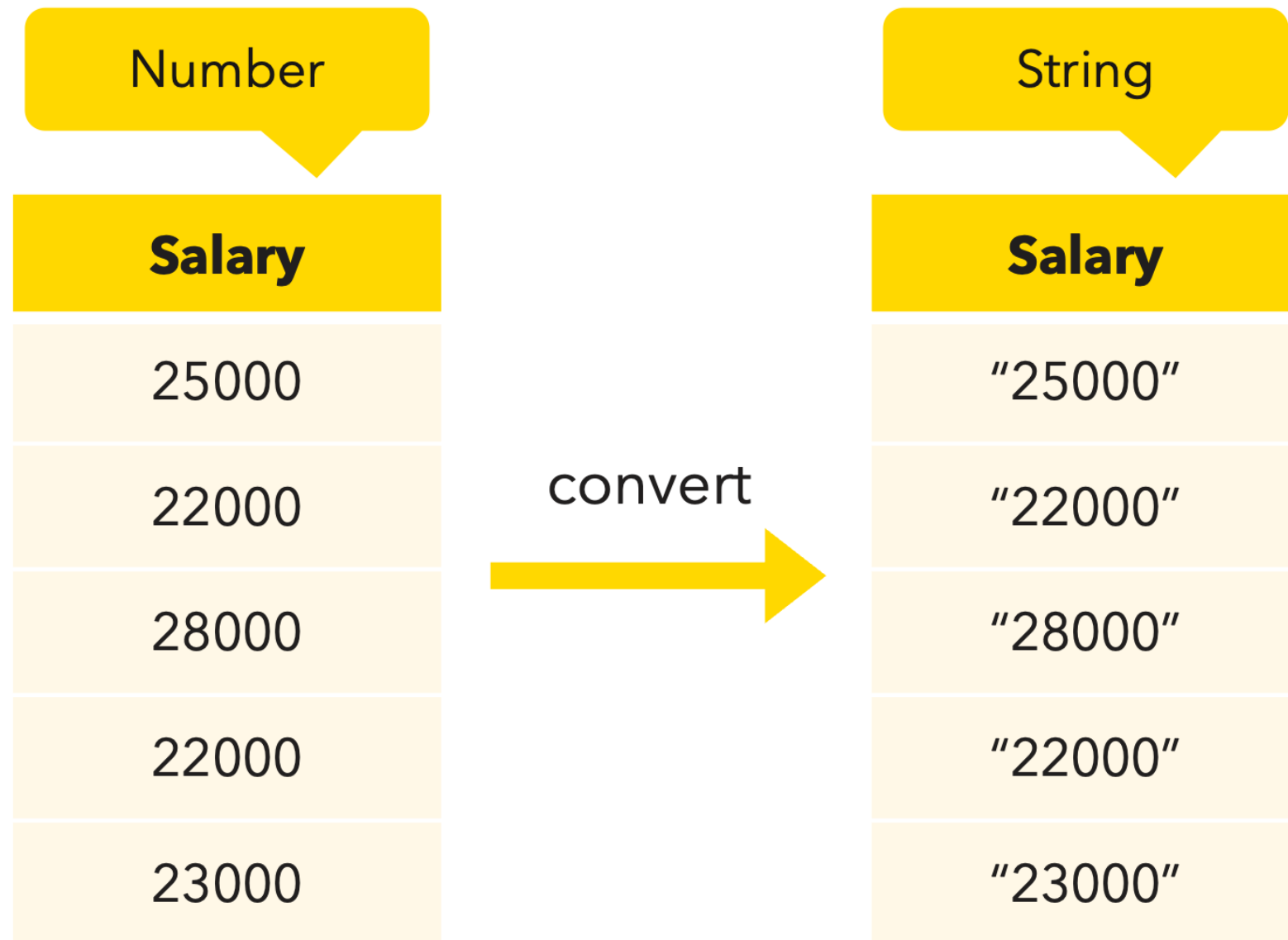
Data types

					String	Number	Date&Time	Boolean
#	Name	Salary	Birthday	Remote				
1	Jenny	25000	12/12/1990	TRUE				
2	Alex	22000	04/03/1998	FALSE				
3	Taylor	28000	11/02/1987	FALSE				
4	Alex	22000	04/03/1998	FALSE				
5	Sam	23000	02/07/1989	TRUE				

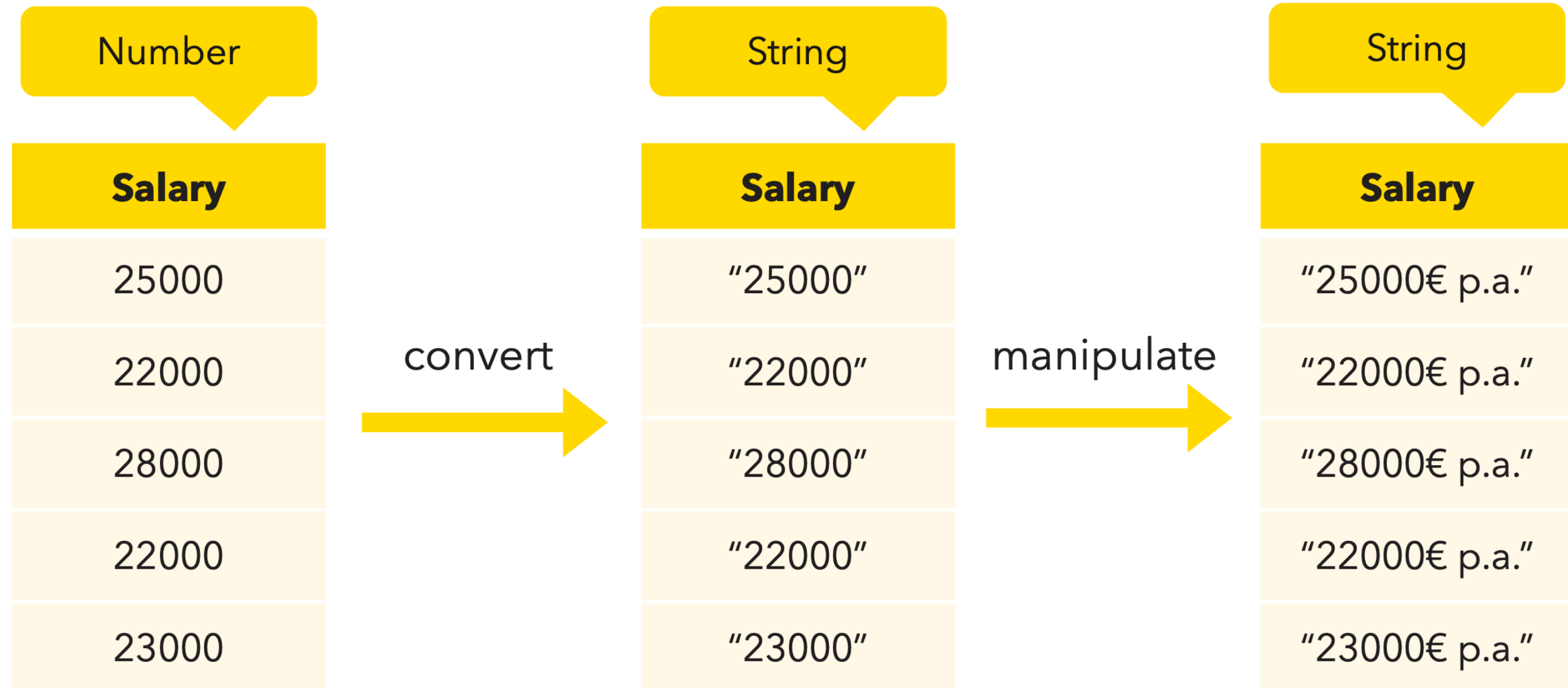
Convert Data Types

Number	
Salary	
	25000
	22000
	28000
	22000
	23000

Convert Data Types



Convert Data Types



Clean strings

Email
jenny.brówn @EXAMPLE.COM
alex.lée @EXAMPLE.COM
taylor.tóñs @EXAMPLE.COM
alex.lée @EXAMPLE.COM
sóphia.kim @EXAMPLE.COM

Clean strings

Email
jenny.brówn @EXAMPLE.COM
alex.lée @EXAMPLE.COM
taylor.tóñs @EXAMPLE.COM
alex.lée @EXAMPLE.COM
sóphia.kim @EXAMPLE.COM

clean



Email
<u>jenny.brown@example.com</u>
<u>alex.lee@example.com</u>
<u>taylor.tons@example.com</u>
<u>alex.lee@example.com</u>
<u>sophia.kim@example.com</u>

Let's practice!

INTRODUCTION TO KNIME

Cleaning data in KNIME Analytics Platform

INTRODUCTION TO KNIME



Full Name
Instructor

Let's practice!

INTRODUCTION TO KNIME